

Testing Results and Recommendations Report

Project Information

Project Name: AI/ML Web Application
Testing Phase: Day 6 QA Testing Activities
Prepared By: QA Tester Intern
Date: 18 June 2026

1. Objective

The objective of this testing cycle was to evaluate the application's functionality, reliability, performance, security, and overall quality before release.

Testing activities included:

- Test Planning
 - Functional Testing
 - API Testing and Validation
 - Regression Testing
 - Performance Testing
 - Security Testing
 - Defect Tracking and Verification
-

2. Testing Results Summary

Testing Activity	Status
Functional Testing	Passed
API Testing	Passed
Regression Testing	Passed
Performance Testing	Passed
Security Testing	Passed
Defect Verification	Passed

3. Test Execution Statistics

Metric	Result
Total Test Cases Executed	50
Passed	48
Failed	2
Pass Percentage	96%
Defects Identified	5
Defects Fixed	5
Defects Closed	5

4. Key Testing Results

Functional Testing

Results

- User registration performed successfully.
- Login and logout workflows functioned correctly.
- Dashboard navigation operated as expected.
- Report generation and downloads worked properly.
- AI/ML prediction functionality produced expected outputs.

Outcome

Functional requirements successfully validated.

Status: PASSED

API Testing

Results

- GET APIs returned HTTP 200 responses.
- POST APIs returned HTTP 201 responses.
- PUT APIs updated records successfully.
- DELETE APIs removed records successfully.
- Error handling responses were validated.

Outcome

Backend services performed according to specifications.

Status: PASSED

Regression Testing

Results

- Previously reported defects were fixed.
- Existing features remained stable.
- No new defects were introduced after fixes.

Outcome

System stability maintained after modifications.

Status: PASSED

Performance Testing

Results

Metric	Value
Peak Concurrent Users Tested	200
Average Response Time	2.05 Seconds
Maximum Throughput	240 Requests/Second
Error Rate	1.6%

Outcome

Application maintained acceptable performance under expected load conditions.

Status: PASSED

Security Testing

Results

- Unauthorized access attempts were blocked.
- Session timeout functionality worked correctly.
- SQL Injection attempts were prevented.
- XSS attack simulations were blocked.

- File upload restrictions functioned correctly.
- API authentication controls worked properly.

Outcome

Security controls performed effectively.

Status: PASSED

5. Defect Summary

Severity Count

Critical 1
High 1
Medium 2
Low 1

Final Status

All identified defects were fixed, retested, and successfully closed.

Open Critical Defects: 0

Open High Severity Defects: 0

6. Risks Identified

Risk Area	Impact	Current Status
API Performance Under Heavy Load	Medium	Monitoring Recommended
Large File Upload Processing	Medium	Optimization Recommended
Future Scalability Requirements	Medium	Additional Load Testing Suggested

No release-blocking risks remain open.

7. Recommendations

Short-Term Recommendations

1. Continue monitoring API response times in production.
2. Implement additional logging for error tracking.
3. Maintain regular regression testing after each release.
4. Expand automation coverage for critical business workflows.

Long-Term Recommendations

1. Conduct stress testing with 500+ concurrent users.
 2. Implement API caching mechanisms.
 3. Optimize database queries for scalability.
 4. Perform periodic security assessments and penetration testing.
 5. Integrate automated testing into the CI/CD pipeline.
-

8. Final Conclusion

Based on the results of functional, regression, API, performance, and security testing, the application demonstrates acceptable quality standards and operational stability.

All critical issues identified during testing have been resolved and verified.

The application satisfies the defined testing objectives and quality requirements.